

01d — What is Hosting

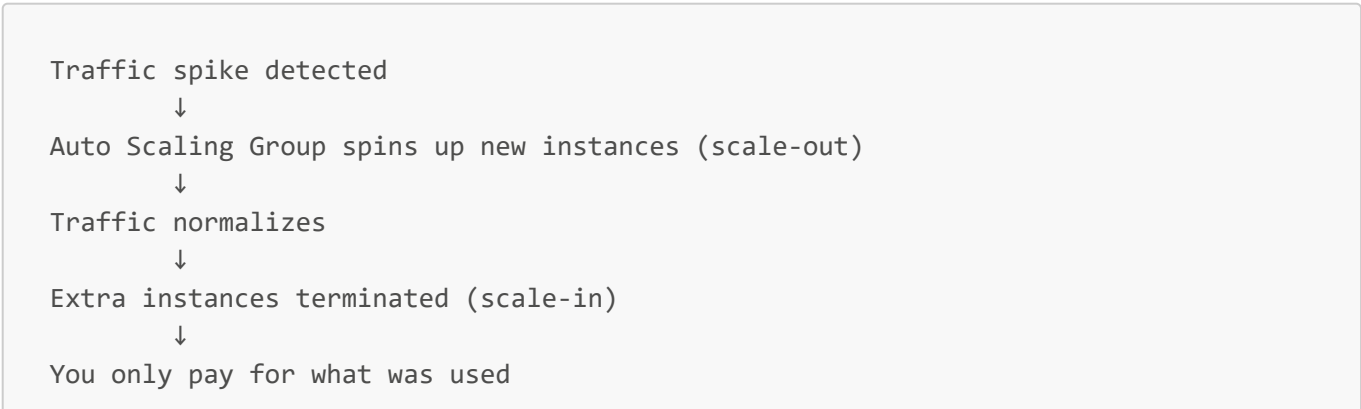
Key Concepts

Term	Definition
Hosting	Renting server infrastructure to run your app on the internet
Shared Hosting	Multiple apps share one server — cheapest, lowest isolation
VPS	Virtual Private Server — isolated slice of a physical server
Dedicated Server	You own the entire physical machine
Cloud Hosting	On-demand servers from AWS/Azure/GCP — pay-as-you-go
PaaS	Platform as a Service — provider manages server, you deploy code
Auto-scaling	Automatically adding/removing server instances based on traffic
Elasticity	Ability to scale up AND back down automatically
Uptime	% of time server is available (99.9% = ~8.7 hrs downtime/year)
SLA	Service Level Agreement — uptime guarantee from provider
Region	Physical data center location (e.g., AWS <code>us-east-1</code> , <code>ap-south-1</code>)

Hosting Types Compared

Type	Isolation	Scalability	Cost	Best For
Shared	Low	None	Cheapest	Blogs, personal sites
VPS	Medium	Manual	Low-mid	Small-medium apps
Dedicated	High	Manual	High	Predictable high traffic
Cloud	High	Automatic	Pay-per-use	Startups to enterprise
PaaS	Managed	Automatic	Mid	Fast deployments

How Auto-Scaling Works



Industry Examples

- **Netflix** — fully on AWS; multiple regions so if one goes down, traffic shifts automatically
- **Airbnb** — migrated from own servers to AWS, reduced ops overhead dramatically
- **Stack Overflow** — dedicated servers; predictable traffic makes it more cost-effective than cloud at their scale

Interview Questions

Q: What is web hosting? A: Renting server infrastructure to run your application so it's accessible over the internet.

Q: What are the types of hosting? A: Shared, VPS, Dedicated, Cloud, and PaaS — each trades isolation and scalability against cost.

Q: Why did the industry move to cloud hosting? A: Elasticity — auto-scale up during traffic spikes, scale back down after. Pay only for what you use.

Q: When would you choose dedicated over cloud? A: When traffic is large but predictable — at scale, owning hardware is cheaper than paying cloud per-hour rates (e.g., Stack Overflow).

Q: What is elasticity in cloud hosting? A: The ability to automatically scale servers up during spikes and back down when traffic drops.